

Project Development Phase

Model Performance Test — Machine Learning

Date: 29 June 2026

Team ID: N/A

Project Name: Network Anomaly Detection

Maximum Marks: 10 Marks

Model Performance Testing:

S.No.	Parameter	Values	Screenshot
1.	Metrics <i>Classification Model</i>	Confusion Matrix: True Normal (TN): 1,738,898 False Positive (FP): 356,159 False Negative (FN): 169,220 True Positive / Anomaly (TP): 256,521 Accuracy Score: 0.79 (79%) Classification Report: Normal — Precision: 0.91 Recall: 0.83 F1: 0.87 Support: 2,095,057 Anomaly — Precision: 0.42 Recall: 0.60 F1: 0.49 Support: 425,741 Macro avg 0.67 / 0.72 / 0.68 2,520,798 Weighted avg 0.83 / 0.79 / 0.81 2,520,798	<i>Confusion matrix heatmap + metrics bar chart from anomaly_dashboard.png (Figure 2, Milestone 7)</i>
2.	Tune the Model	Hyperparameter Tuning: Algorithm: Isolation Forest (scikit-learn IsolationForest) n_estimators = 100 (100 trees in the ensemble) contamination = 0.17 (calibrated from EDA: 16.9% attack ratio, vs scikit-learn default of 0.1) random_state = 42 (reproducibility) n_jobs = -1 (all CPU cores) Validation Method: Training set: 2,095,057 rows of normal (BENIGN) traffic only. Test set: 2,520,798 rows (full dataset — normal + all attack types).	<i>Code snippet showing IsolationForest(...) constructor with above parameters</i>

S.No.	Parameter	Values	Screenshot
		Ground-truth labels (Label_binary) held out for post-hoc evaluation only; never used during training, preserving the unsupervised nature of the model. Evaluation metrics: sklearn classification_report + confusion_matrix.	